# Java 21 Virtual Threads & Structured Concurrency • INSTRUCTOR ANSWER KEY

Course: Java 21.0.6 • College: VIT • Student Name: _____

### Q1. In Java 21, what key advantage do Virtual Threads offer over traditional Platform Threads?

**[CORRECT]** A) High throughput for I/O-bound tasks with minimal OS thread overhead

B) Direct acceleration using GPU hardware cores

C) Elimination of garbage collection pauses entirely

D) Automatic conversion of all blocking code into C++ binaries

**Key: A** — Virtual threads are lightweight threads managed by the JVM that dramatically reduce memory and context-switching overhead for I/O bound concurrency.

**Distractor Analysis:**

- **Option B:** Virtual threads run on the CPU via carrier threads and do not interact with GPU hardware.
- **Option C:** Garbage collection behavior is unchanged; objects allocated by virtual threads are still subject to GC.
- **Option D:** Virtual threads remain standard JVM bytecode and are not converted to C++ binaries.

### Q2. Which compiler flag in modern javac treats terminally deprecated APIs as build failures?

**[CORRECT]** A) -Xlint:deprecation,removal

B) -Wno-all

C) --ignore-warnings

D) -source 1.8

**Key: A** — Specifying -Xlint:deprecation,removal enables strict deprecation audits where APIs marked for removal trigger fatal warnings.

**Distractor Analysis:**

- **Option B:** -Wno-all is a gcc flag, not a javac option.
- **Option C:** --ignore-warnings is invalid javac syntax.
- **Option D:** -source 1.8 downgrades language syntax to legacy Java 8.

### Q3. Why is deterministic output normalization critical in automated code verification for Java?

**[CORRECT]** A) It masks memory addresses, timestamps, and thread IDs so identical logic yields identical results

B) It rewrites slow algorithms into faster ones automatically

C) It disables unit testing to make test suites run faster

D) It converts console output into binary machine code

**Key: A** — Normalizing transient data like pointers (0x7ffe...) and timestamps ensures reproducible regression testing.

**Distractor Analysis:**

- **Option B:** Normalization does not alter algorithmic complexity.
- **Option C:** Testing remains active; only the output stream is sanitized.
- **Option D:** Output strings remain text; they are not converted to machine code.

### Q4. What is the primary objective of mutation testing in exercise validation?

**[CORRECT]** A) Proving that student-facing test suites successfully fail when code logic is intentionally mutated/broken

B) Randomly deleting student files to test backup recovery

C) Speeding up Docker container build times

D) Encrypting source code with AES-256

**Key: A** — Mutation testing validates that tests are robust enough to catch real logical faults (killing mutants).

**Distractor Analysis:**

- **Option B:** Mutation testing operates in memory and does not delete files.
- **Option C:** Mutation testing evaluates test quality, not container build speed.
- **Option D:** It has no connection to cryptographic encryption.

### Q5. In Java, what is the main consequence of executing untrusted code without container sandbox isolation?

**[CORRECT]** A) Direct host filesystem compromise, resource exhaustion, and external network leaks

B) Syntax highlighting colors changing randomly

C) The compiler running 100x faster than expected

D) PowerPoint slide fonts changing to Comic Sans

**Key: A** — Unsandboxed execution allows hostile code to escape into the host OS, read private files, or launch network attacks.

**Distractor Analysis:**

- **Option B:** Syntax highlighting is visual styling and unrelated to process isolation.
- **Option C:** Unsandboxed execution does not improve compiler speed.
- **Option D:** Font styles in presentations are completely decoupled from OS sandboxing.

### Q6. When configuring container limits for isolated verification, why is '--network none' strictly enforced?

**[CORRECT]** A) To prevent code from downloading malicious payloads, leaking credentials, or contacting external servers

B) Because Docker containers cannot support networking on Linux

C) To decrease the size of Docker image layers on disk

D) To disable JSON serialization in the API

**Key: A** — --network none guarantees complete network isolation, ensuring runs remain offline and secure.

**Distractor Analysis:**

- **Option B:** Docker supports full bridge and host networking; disabling it is a security decision.
- **Option C:** Network configuration does not affect image layer size.
- **Option D:** JSON serialization is an application-level data format independent of network stacks.

### Q7. What is the recommended approach for handling deprecation warnings in current production Java?

**[CORRECT]** A) Treat warnings as errors in CI/CD pipelines and migrate to the approved modern replacements

B) Pin older runtime versions indefinitely to avoid updating

C) Turn off compiler error logging across the entire company

D) Use reflection to bypass deprecated method signatures

**Key: A** — Treating deprecations as errors in CI ensures codebases stay healthy and avoids breaking when APIs are removed.

**Distractor Analysis:**

- **Option B:** Pinning ancient versions creates massive technical debt and severe security vulnerabilities.
- **Option C:** Disabling logging blinds teams to fatal production failures.
- **Option D:** Reflection does not eliminate deprecation and adds runtime fragility.

### Q8. Which practice is recommended when engineering scalable Java applications for Java 21 Virtual Threads & Structured Concurrency?

**[CORRECT]** A) Use verified idioms, explicit types, and automated unit test assertions

B) Suppress all compiler and deprecation warnings to simplify logs

C) Rely entirely on global mutable state to avoid passing parameters

D) Execute all database calls synchronously inside UI or main thread loops

**Key: A** — Verified engineering practices in Java mandate explicit typing, zero warning tolerance, and automated test suites.

**Distractor Analysis:**

- **Option B:** Suppressing compiler warnings masks impending API removals and security regressions.
- **Option C:** Global mutable state causes unpredictable side effects and race conditions.
- **Option D:** Synchronous database queries on main threads cause severe UI freezing and I/O bottlenecks.

## Q9. What is the exact output produced by executing this Java code?

```
public class Solution {
public static void main(String[] args) {
int a = 5;
int b = a++ * 2;
System.out.println(a + "," + b);
}
}
```

A) Compilation Error / Syntax Warning

**[CORRECT]** B) 6,10

C) null

D) Runtime Exception: IndexOutOfBounds

**Key: B** — Executing this snippet in the Java sandbox yields: '6,10'.

**Distractor Analysis:**

- **Option A:** The code is syntactically valid and compiles cleanly under the pinned Java runtime.
- **Option C:** The evaluation resolves to a concrete computed value rather than null.
- **Option D:** All access boundaries are respected, producing no runtime exceptions.

## Q10. What is the exact output produced by executing this Java code?

```
public class Solution {
public static void main(String[] args) {
String s = "Java 21";
System.out.println(s.substring(5) + "-" + s.length());
}
}
```

A) Compilation Error / Syntax Warning

**[CORRECT]** B) 21-7

C) null

D) Runtime Exception: IndexOutOfBounds

**Key: B** — Executing this snippet in the Java sandbox yields: '21-7'.

**Distractor Analysis:**

- **Option A:** The code is syntactically valid and compiles cleanly under the pinned Java runtime.
- **Option C:** The evaluation resolves to a concrete computed value rather than null.
- **Option D:** All access boundaries are respected, producing no runtime exceptions.